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Introduction

1.1 Probability Spaces

Definition 1.1.1 ► Probability Spaces

A **probability space** is a measure space $(\Omega, \mathcal{F}, P)$ with

$$P : \mathcal{F} \rightarrow [0, 1]$$

Example 1.1.2 ► Discrete probability spaces

Let Ω be a countable set . Let $\mathcal{F}$ be the collection of all subsets of Ω . Let

$$P(A) = \sum_{\omega \in A} p(\omega), \quad \text{where } p(\omega) \geq 0 \text{ and } \sum_{\omega \in \Omega} p(\omega) = 1$$

Example 1.1.3 ► Measures on the real line.

Measures on $(\mathbf{R}, \mathcal{R})$ are defined by giving probability a **Stieltjes measure function** with the following properties:

- (i) F is nondecreasing.
- (ii) F is right continuous, that is, $\lim_{y \downarrow x} F(y) = F(x)$.

Theorem 1.1.4

Associated with each Stieltjes measure function F there is a unique measure μ on $(\mathbf{R}, \mathcal{R})$ with $\mu((a, b]) = F(b) - F(a)$

$$\mu((a, b]) = F(b) - F(a) \tag{1.1}$$

When $F(x) = x$ the resulting measure is called **Lebesgue measure**

Remark.

The proof is long and winding. We will show the main idea following.

Definition 1.1.5 ▶ Semialgebra

A collection $\mathcal{S}$ of sets is said to be a **semialgebra** if (i) it is closed under intersection, that is, $S, T \in \mathcal{S}$ implies $S \cap T \in \mathcal{S}$, and (ii) if $S \in \mathcal{S}$ then S^c is a finite disjoint union of sets in $\mathcal{S}$.

Example 1.1.6

$S_d =$ the empty set plus all sets of the form

$$(a_1, b_1] \times \cdots \times (a_d, b_d] \subset \mathbb{R}^d \quad \text{where } -\infty \leq a_i < b_i \leq \infty$$

Then S_d is a semialgebra.

Remark.

The above Stieljes measure function gives the values of μ on the semialgebra.

Definition 1.1.7 ▶ Algebra

A collection $\mathcal{A}$ of subsets of Ω is called an **algebra** (or **field**) if $A, B \in \mathcal{A}$ implies A^c and $A \cup B$ are in $\mathcal{A}$.

Remark.

Since $A \cap B = (A^c \cup B^c)^c$, it follows that $A \cap B \in \mathcal{A}$. Obviously a σ -algebra is an algebra.

Lemma 1.1.8

If $\mathcal{S}$ is a semialgebra, then $\bar{\mathcal{S}} = \{\text{finite disjoint unions of sets in } \mathcal{S}\}$ is an algebra, called the **algebra generated by $\mathcal{S}$** .

Remark.

Given a set function μ on $\mathcal{S}$, we can extend it to $\bar{\mathcal{S}}$ by

$$\mu\left(+_{i=1}^n A_i\right) = \sum_{i=1}^n \mu(A_i)$$

Proof. Suppose $A = +_i S_i$ and $B = +_j T_j$, where $+$ denotes disjoint union and we assume the index sets are finite. Then $A \cup B = +_{i,j} S_i \cap T_j$, where $S_j \cap T_j \in \mathcal{S}$

by definition. Thus $\bar{\mathcal{S}}$ is closed under intersection. As for complements, if $A = +_i \mathcal{S}_i$, then $A^c = \cap_i \mathcal{S}_i^c$, where $\mathcal{S}_i^c$ is a finite disjoint union of sets in $\mathcal{S}$, thus is in $\bar{\mathcal{S}}$. Since $\bar{\mathcal{S}}$ is closed under intersection, then $A^c \in \bar{\mathcal{S}}$ by induction. $\square$

Definition 1.1.9 ► measure on an algebra

By a **measure on an algebra** $\mathcal{A}$, we mean a set function μ with

- (i) $\mu(A) \geq \mu(\emptyset) = 0$ for all $A \in \mathcal{A}$, and
- (ii) if $A_i \in \mathcal{A}$ are **disjoint** and **their union is in** $\mathcal{A}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Definition 1.1.10 ► σ -finite

μ is said to be **σ -finite** if there is a sequence of sets $A_n \in \mathcal{A}$ so that $\mu(A_n) < \infty$ and $\cup_n A_n = \Omega$.

Remark.

Letting $A'_1 = A_1$ and for $n \geq 2$,

$$A'_n = \bigcup_{m=1}^n A_m \quad \text{or} \quad A'_n = A_n \cap \left(\bigcap_{m=1}^{n-1} A_m^c \right) \in \mathcal{A}$$

we can without loss of generality assume that $A_n \uparrow \Omega$ or the A_n are disjoint.

Theorem 1.1.11

Let $\mathcal{S}$ be a semialgebra and let μ defined on $\mathcal{S}$ have $\mu(\emptyset) = 0$. Suppose (i) if $S \in \mathcal{S}$ is a finite disjoint union of sets $S_i \in \mathcal{S}$ then $\mu(S) = \sum_i \mu(S_i)$, and (ii) if $S_i, S \in \mathcal{S}$ with $S = +_{i \geq 1} S_i$ then $\mu(S) \leq \sum_{i \geq 1} \mu(S_i)$. Then μ has a unique extension $\bar{\mu}$ that is a measure on $\bar{\mathcal{S}}$ the algebra generated by $\mathcal{S}$. If $\bar{\mu}$ is sigma-finite then there is a unique extension ν that is a measure on $\sigma(\mathcal{S})$.

Lemma 1.1.12

Suppose only that (i) holds.

1. If $A, B_i \in \bar{\mathcal{S}}$ with $A = +_{i=1}^n B_i$, then $\bar{\mu}(A) = \sum_i \bar{\mu}(B_i)$.

2. If $A, B_i \in \bar{\mathcal{S}}$ with $A \subseteq \bigcup_{i=1}^n B_i$ then $\bar{\mu}(A) \leq \sum_i \bar{\mu}(B_i)$

Measures on $\mathbb{R}^d$

1.1.1 Exercice

Exercise 1.1.13

A set $A \subset \{1, 2, \dots\}$ is said to have *asymptotic density* θ if

$$\lim_{n \rightarrow \infty} |A \cap \{1, 2, \dots, n\}|/n = \theta$$

Let $\mathcal{A}$ be the collection of sets for which the asymptotic density exists. Is $\mathcal{A}$ a σ -algebra? an algebra?

Proof. Suppose that $A \in \mathcal{A}$ has asymptotic density θ .

$$|A \cap \{1, 2, \dots, n\}|/n + |A^c \cap \{1, 2, \dots, n\}|/n = 1$$

Take the limit of $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} |A \cap \{1, 2, \dots, n\}|/n + \lim_{n \rightarrow \infty} |A^c \cap \{1, \dots, n\}|/n = 1$$

which implies that A^c has the asymptotic density $1 - \theta$.

Let $A = \{2k\}_{k=1}^{\infty}$. Let B contained all the even numbers among $[2^{2m}, 2^{2m+1} - 1]$, and contained all the odd numbers among $[2^{2m+1}, 2^{2m+2} - 1]$. Since there are as many even numbers as odd numbers among $[2^k, 2^k - 1]$. It is not hard to see B has asymptotic density $\frac{1}{2}$. Then

$$\begin{aligned} \lim_{m \rightarrow \infty} |(A \cap B) \cap \{1, 2, \dots, 2^{2m+1}\}|/2^{2m+1} &= \lim_{m \rightarrow \infty} \sum_{k=0}^m \frac{1}{2} (2^{2k+1} - 2^{2k})/2^{2m+1} \\ &= \lim_{m \rightarrow \infty} \sum_{k=0}^m 4^{k-m-1} \\ &= \lim_{m \rightarrow \infty} 4^{-m-1} \frac{1 - 4^{m+1}}{1 - 4} = \frac{1}{3} \end{aligned}$$

$$\lim_{m \rightarrow \infty} |(A \cap B) \cap \{1, 2, \dots, 2^{2m+2}\}| / 2^{2m+2} = \lim_{m \rightarrow \infty} \sum_{k=0}^m \frac{1}{2} (2^{2m+1} - 2^{2m}) / 2^{2m+2} = \frac{1}{6}$$

which implies that $A \cap B$ is not in $\mathcal{A}$. But $\mathcal{A}$ is closed under complements, $\mathcal{A}$ is not closed under finite union. Thus $\mathcal{A}$ is not an algebra, then is neither a σ -algebra. $\square$

1.2 Distributions

Definition 1.2.1

A real-valued function X defined on Ω is said to be a **random variable** if for every Borel set $B \subseteq \mathbb{R}$ we have $X^{-1}(B) \in \mathcal{F}$. That is a real valued measurable function on $\mathcal{F}$. We write $X \in \mathcal{F}$.

Example 1.2.2

1. If Ω is a discrete probability space, then any function $X : \Omega \rightarrow \mathbb{R}$ is a random variable.
2. For any $A \in \mathcal{F}$, the **indicator function** of A is:

$$1_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

Definition 1.2.3 ► distribution

- If X is a random variable, then X induces a probability measure on $\mathbb{R}$ called **distribution** by setting $\mu(A) = P(X \in A)$ for Borel sets A .
- The distribution of a random variable X is usually described by giving its **distribution function**,

$$F(x) = P(X \leq x)$$

Remark.

The right-hand side can be written as $P(X^{-1}(A))$. In words, we pull $A \in \mathbb{R}$ back to $X^{-1}(A) \in \mathcal{F}$ and then take P of that set.

Proof. If the A_i are disjoint,

$$\begin{aligned}\mu\left(\bigcup_i A_i\right) &= P(X \in A) \\ &= P\left(\bigcup_i \{X \in A_i\}\right) \\ &= \sum_i P(X \in A_i) = \sum_i \mu(A_i)\end{aligned}$$

□

Theorem 1.2.4

Any distribution function F has the following properties:

- (i) F is **nondecreasing**.
- (ii) $\lim_{x \rightarrow \infty} F(x) = 1$, $\lim_{x \rightarrow -\infty} F(x) = 0$.
- (iii) F is **right continuous**, that is, $\lim_{y \downarrow x} F(y) = F(x)$.
- (iv) If $F(x-) = \lim_{y \uparrow x} F(y)$ then $F(x-) = P(X < x)$.
- (v) $P(X = x) = F(x) - F(x-)$.

Theorem 1.2.5

If F satisfies (i), (ii), and (iii) in the above theorem, then it is the distribution function of some random variable.

Remark.

Even though F may not be 1-1 and onto, we will call X the inverse of F and denote by F^{-1} .

Proof. Let P be Lebesgue measure, Note that

$$F(x) = P(\omega : \omega \leq F(x))$$

And if X is a random variable which we need, then

$$F(x) = P(\omega : X(\omega) \leq x)$$

Thus we only need to show that

$$\{\omega : X(\omega) \leq x\} = \{\omega : \omega \leq F(x)\}$$

holds for some X . If F is a bijection, we may define

$$X(\omega) = F^{-1}(\omega)$$

It motivates us to define

$$X(\omega) = \sup \{y : F(y) < \omega\}$$

in a general case. To check that it is just what we need, we observe that if $\omega \leq F(x)$, and if $x \leq X(\omega)$, then $F(x) < \omega$, which is a contradiction. Thus $x > X(\omega)$. On the other hand if $\omega > F(x)$, then since F is right continuous, there is an $\varepsilon > 0$ so that $F(x + \varepsilon) < \omega$, and $X(\omega) \geq x + \varepsilon > x$. $\square$

Definition 1.2.6 ► Equal in Distribution

- If X and Y induce the same distribution μ on $(\mathbb{R}, \mathcal{R})$, we say X and Y are **equal in distribution**, written by

$$X =_d Y$$

- When the distribution function $F(x) = P(X \leq x)$ has the form

$$F(x) = \int_{-\infty}^x f(y) \, dy$$

We say that X has **density function** f . We write $f_X(x)$.

Remark.

- X and Y are equal in distribution, iff X and Y have the same distribution function, that is, $P(X \leq x) = P(Y \leq x)$ for all x .
- In remembering formulas, it is often useful to think of $f(x)$ as being $P(X = x)$ although

$$P(X = x) = F(x) - F(x-) = \lim_{\varepsilon \rightarrow 0} \int_{x-\varepsilon}^{x+\varepsilon} f(y) \, dy = 0$$

Example 1.2.7 ▶ Uniform distribution on (0,1)

$f(x) = 1$ for $x \in (0, 1)$ and 0 otherwise. Distribution function:

$$F(x) = \begin{cases} 0 & x \leq 0 \\ x & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

Example 1.2.8 ▶ Exponential distribution with rate λ

$f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and 0 otherwise. Distribution function:

$$F(x) = \begin{cases} 0 & x \leq 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

Example 1.2.9 ▶ Standard normal distribution

$$f(x) = (2\pi)^{-1/2} \exp(-x^2/2)$$

Remark.

In this case, there is no closed-form expression for $F(x)$, but we have the following bounds that are useful for large x :

Theorem 1.2.10 ▶ For $x > 0$

$$(x^{-1} - x^{-3}) \exp(-x^2/2) \leq \int_x^\infty \exp(-y^2/2) dy \leq x^{-1} \exp(-x^2/2)$$

1.3 Random Variables

Definition 1.3.1

A function $X : \Omega \rightarrow S$ is said to be a **measurable map** from $(\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ if

$$X^{-1}(B) \equiv \{\omega : X(\omega) \in B\} \in \mathcal{F} \quad \text{for all } B \in \mathcal{S}$$

If $(S, \mathcal{S}) = (\mathbb{R}^d, \mathcal{R}^d)$ and $d > 1$, then X is called a **random vector**. If $d = 1$, X is called a **random variable**, ro r.v. for short.

Theorem 1.3.2

If $\{\omega : X(\omega) \in A\} \in \mathcal{F}$ for all $A \in \mathcal{A}$ and $\mathcal{A}$ **generates** $\mathcal{S}$, then X is measurable.

Definition 1.3.3

Suppose that X is a measurable map. Then we define

$$\sigma(X) := \{\{X \in B\} : B \in \mathcal{S}\}$$

as the **σ -field generated by X** .

Remark.

where we write $\{X \in B\}$ as shorthand for $\{\omega : X(\omega) \in B\}$.

Example 1.3.4

If $(S, \mathcal{S}) = (\mathbb{R}, \mathcal{R})$, then possible choices of $\mathcal{A}$ in 1.3.3 are $\{(-\infty, x] : x \in \mathbb{R}\}$ or $\{(-\infty, x) : x \in \mathbb{Q}\}$.

Example 1.3.5

If $(S, \mathcal{S}) = (\mathbb{R}^d, \mathcal{R}^d)$, a useful choice of $\mathcal{A}$ is

$$\{(a_1, b_1) \times \cdots \times (a_d, b_d) : -\infty < a_i < b_i < \infty\}$$

Theorem 1.3.6

If $X : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ and $f : (S, \mathcal{S}) \rightarrow (T, \mathcal{T})$ are measurable maps, then $f(X)$ is a measurable map from $(\Omega, \mathcal{F})$ to $(T, \mathcal{T})$.

Theorem 1.3.7

If $X_1, \dots, X_n$ are random variables and $f : (\mathbb{R}^n, \mathcal{R}^n) \rightarrow (\mathbb{R}, \mathcal{R})$ is measurable, then $f(X_1, \dots, X_n)$ is a random variable.

Theorem 1.3.8

If $X_1, \dots, X_n$ are random variables then $X_1 + \cdots + X_n$ is a random variable.

Theorem 1.3.9

f $X_1, X_2, \dots$ are random variables then so are

$$\inf_n X_n \quad \sup_n X_n \quad \limsup_n X_n \quad \liminf_n X_n$$

Exercise**Exercise 1.3.10**

Show that if $\mathcal{A}$ generates $\mathcal{S}$, then $X^{-1}(\mathcal{A}) \equiv \{\{X \in A\} : A \in \mathcal{A}\}$ generates $\sigma(X) = \{\{X \in B\} : B \in \mathcal{S}\}$

Proof. First, we show that

$$\sigma(X^{-1}(\mathcal{A})) \subseteq X^{-1}(\mathcal{S})$$

For which, we only need to show that

1. $X^{-1}(\mathcal{S})$ is a σ -algebra;
 2. $X^{-1}(\mathcal{A}) \subseteq X^{-1}(\mathcal{S})$.
2. is obvious from $\mathcal{A} \subseteq \mathcal{S}$. Since $\mathcal{S}$ is a σ -algebra, and the complementation, union are kept under preimage, then $X^{-1}(\mathcal{S})$ is a σ -algebra.

Next, we show that

$$X^{-1}(\mathcal{S}) \subseteq \sigma(X^{-1}(\mathcal{A}))$$

We define

$$\mathcal{C} := \{B \in \mathcal{S} : X^{-1}(B) \in \sigma(X^{-1}(\mathcal{A}))\}$$

It is not hard to see that $\mathcal{C}$ is a σ -algebra. Once we show that $\mathcal{A} \subseteq \mathcal{C}$, we have $\mathcal{S} \subseteq \mathcal{C}$, then it follows that $X^{-1}(\mathcal{S}) \subseteq \sigma(X^{-1}(\mathcal{A}))$. For that, we take $A \in \mathcal{A}$, since $X^{-1}(A) \in X^{-1}(\mathcal{A}) \subseteq \sigma(X^{-1}(\mathcal{A}))$, $A \in \mathcal{C}$. Thus we completes the proof. $\square$

1.4 Expected Value

Definition 1.4.1 ► Expected Value

- If $X \geq 0$ is a random variable on $(\Omega, \mathcal{F}, P)$, then we define its **expected value** to be

$$EX = \int X \, dP$$

We declare that EX **exists** and set

$$EX = EX^+ - EX^-$$

whenever the subtraction makes sense.

Remark.

EX is often called the **mean** of X and denoted by μ .

Theorem 1.4.2

Suppose $X, Y \geq 0$ or $E|X|, E|Y| < \infty$.

- $E(X + Y) = EX + EY$.
- $E(aX + b) = aE(X) + b$ for any real numbers a, b .
- If $X \geq Y$ then $EX \geq EY$.

1.4.1 Inequalities

Theorem 1.4.3 ► Jensen

Suppose φ is **convex**. Then

$$E(\varphi(X)) \geq \varphi(EX)$$

provided both expectations exist, that is, $E|X|$ and $E|\varphi(X)| < \infty$

Theorem 1.4.4 ► Holder

If $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$, then

$$E|XY| \leq \|X\|_p \|Y\|_q$$

Here $\|X\|_r = (E|X|^r)^{\frac{1}{r}}$ for $r \in [1, \infty)$; $\|X\|_\infty = \inf\{M : P(|X| > M) = 0\}$

1.4.2 Integration to the Limit

Theorem 1.4.5 ▶ Fatou's lemma

If $X_n \geq 0$ then

$$\liminf_{n \rightarrow \infty} EX_n \geq E\left(\liminf_{n \rightarrow \infty} X_n\right)$$

Theorem 1.4.6 ▶ Monotone convergence theorem

If $0 \leq X_n \uparrow X$ then $EX_n \uparrow EX$.

Theorem 1.4.7 ▶ Dominated convergence theorem

If $X_n \rightarrow X$ a.s., $|X_n| \leq Y$ for all n , and $EY < \infty$, then $EX_n \rightarrow EX$.

1.4.3 Computing Expected Values

Theorem 1.4.8 ▶ Change of variables formula

Let X be a random element of $(S, \mathcal{S})$ with distribution μ , that is, $\mu(A) = P(X \in A)$. If f is a measurable function from $(S, \mathcal{S})$ to $(\mathbb{R}, \mathcal{R})$ so that $f \geq 0$ or $E|f(X)| < \infty$, then

$$Ef(X) = \int_S f \, d\mu =: \int_S f(y) \mu(dy)$$

Definition 1.4.9

- If k is a positive integer, then EX^k is called the ***kth moment*** of X . The first moment EX is usually called the ***mean*** and denoted by μ .
- If $EX^2 < \infty$, then the ***variance*** of X is defined to be

$$\begin{aligned} \text{var}(X) &= E(X - \mu)^2 \\ &= EX^2 - \mu^2 \end{aligned}$$

Proposition 1.4.10

- $\text{var}(X) \leq EX^2$
- $\text{var}(aX + b) = a^2\text{var}(X)$

Example 1.4.11

If X has the *exponential distribution* with rate 1, then

$$EX^k = \int_0^{\infty} x^k e^{-x} dx = k!$$

Proof.

$$\begin{aligned} EX^k &= \int_S y^k d\mu = \int_S y^k dF = \int_{\mathbb{R}} x^k f(x) dx = \int_0^{\infty} x^k e^{-x} dx \\ \int_0^{\infty} x^k e^{-x} dx &= - \int_0^{\infty} x^k de^{-x} = -[e^{-x} x^k]_0^{\infty} + k \int_0^{\infty} x^{k-1} e^{-x} dx \\ &= k \int_0^{\infty} x^{k-1} e^{-x} dx \end{aligned}$$

Then it follows from induction. □

Important Distribution and Their Calculations

2.1 Discrete Distributions

Proposition 2.1.1 ► Bernoulli Distribution

Let $p \in [0, 1]$. If a random variable X takes values in $\{0, 1\}$ and satisfies $P(X = 1) = p, P(X = 0) = 1 - p$, then X is said to follow a ***Bernoulli distribution*** with parameter p , denoted as $X \sim \text{Bernoulli}(p)$. In this case:

1. The probability mass function is

$$P(X = k) = p^k(1 - p)^{1-k}, \quad k \in \{0, 1\};$$

2. $E[X] = p$;
3. $\text{Var}(X) = p(1 - p)$;
4. The moment generating function is

$$M_X(t) = 1 - p + pe^t.$$

Proposition 2.1.2 ► Binomial Distribution

Let $n \in \mathbb{N}^+, p \in [0, 1]$. If a random variable X represents the number of successes in n independent repeated Bernoulli trials, then X is said to follow a ***binomial distribution*** with parameters (n, p) , denoted as $X \sim B(n, p)$. In this case:

1. The probability mass function is

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n;$$

2. $E[X] = np$;
3. $\text{Var}(X) = np(1 - p)$;

4. The moment generating function is

$$M_X(t) = (1 - p + pe^t)^n;$$

5. If $X \sim B(n_1, p)$, $Y \sim B(n_2, p)$ and X, Y are independent, then

$$X + Y \sim B(n_1 + n_2, p).$$

Proposition 2.1.3 ► Geometric Distribution

Let $p \in (0, 1]$. If a random variable X represents the number of trials needed to obtain the first success in independent repeated Bernoulli trials, then X is said to follow a **geometric distribution** with parameter p . In this case:

1. The probability mass function is

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots;$$

2.

$$E[X] = \frac{1}{p};$$

3.

$$\text{Var}(X) = \frac{1 - p}{p^2};$$

4. Memoryless property: for any $m, n \in \mathbb{N}^+$, we have

$$P(X > m + n | X > m) = P(X > n);$$

5. The moment generating function is

$$M_X(t) = \frac{pe^t}{1 - (1 - p)e^t}, \quad t < -\ln(1 - p).$$

Proof.

$$\begin{aligned}
 P(X > m + n | X > m) &= \frac{P(X > m + n)}{P(X > m)} = \frac{\sum_{i=m+n+1}^{\infty} (1-p)^{i-1} p}{\sum_{i=m+1}^{\infty} (1-p)^{i-1} p} = \frac{\frac{(1-p)^{m+n}}{p}}{\frac{(1-p)^m}{p}} \\
 &= \frac{(1-p)^n}{p} p \\
 &= \sum_{i=n+1}^{\infty} (1-p)^{i-1} p \\
 &= P(X > n)
 \end{aligned}$$

□

Proposition 2.1.4 ► Negative Binomial Distribution

Let $r \in \mathbb{N}^+$, $p \in (0, 1]$. If a random variable X represents the number of trials needed to obtain the r -th success in independent repeated Bernoulli trials, then X is said to follow a **negative binomial distribution** with parameters (r, p) , denoted as $X \sim NB(r, p)$. In this case:

1. The probability mass function is

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, \dots;$$

- 2.

$$E[X] = \frac{r}{p};$$

- 3.

$$\text{Var}(X) = \frac{r(1-p)}{p^2};$$

4. When $r = 1$, the negative binomial distribution reduces to the geometric distribution;
5. If $X \sim NB(r_1, p)$, $Y \sim NB(r_2, p)$ and X, Y are independent, then

$$X + Y \sim NB(r_1 + r_2, p).$$

Proposition 2.1.5 ▶ Poisson Distribution

Let $\lambda > 0$. If a random variable X takes values in $\mathbb{N} \cup \{0\}$ with probability mass function

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!},$$

then X is said to follow a **Poisson distribution** with parameter λ , denoted as $X \sim \text{Poisson}(\lambda)$. In this case:

1. $E[X] = \lambda$;
2. $\text{Var}(X) = \lambda$;
3. The moment generating function is

$$M_X(t) = e^{\lambda(e^t - 1)};$$

4. If $X \sim \text{Poisson}(\lambda_1)$, $Y \sim \text{Poisson}(\lambda_2)$ and X, Y are independent, then

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2).$$

Proposition 2.1.6 ▶ Poisson Limit Theorem

Let $X_n \sim B(n, p_n)$, where $np_n \rightarrow \lambda > 0$ as $n \rightarrow \infty$. Then for any $k \in \mathbb{N} \cup \{0\}$, we have

$$\lim_{n \rightarrow \infty} P(X_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

Proposition 2.1.7 ▶ Hypergeometric Distribution

Let $N, M, n \in \mathbb{N}^+$ with $M \leq N, n \leq N$. If a random variable X represents the number of success elements when drawing n elements without replacement from a population containing M success elements and $N - M$ failure elements, then X is said to follow a **hypergeometric distribution** with parameters (N, M, n) , denoted as $X \sim H(N, M, n)$. In this case:

1. The probability mass function is

$$P(X = k) = \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}}, \quad k = \max(0, n - N + M), \dots, \min(n, M);$$

2.

$$E[X] = n \frac{M}{N};$$

3.

$$\text{Var}(X) = n \frac{M}{N} \cdot \frac{N-M}{N} \cdot \frac{N-n}{N-1};$$

4. As $N \rightarrow \infty$ and $\frac{M}{N} \rightarrow p$, the hypergeometric distribution converges in distribution to the binomial distribution $B(n, p)$.

2.2 Continuous Distribution

Proposition 2.2.1 ▶ Uniform Distribution

Let $a, b \in \mathbb{R}$ with $a < b$. If a random variable X has a probability density function (PDF) that is constant over the interval $[a, b]$, then X is said to follow a **uniform distribution** on $[a, b]$, denoted as $X \sim U(a, b)$. In this case:

1. The probability density function is

$$\begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise.} \end{cases}$$

2. $E[X] = \frac{a+b}{2};$

3. $\text{Var}(X) = \frac{(b-a)^2}{12};$

4. The moment generating function is $M_X(t) = \frac{e^{tb} - e^{ta}}{t(b-a)}, \quad \text{for } t \neq 0.$

Proposition 2.2.2 ▶ Exponential Distribution

Let $\lambda > 0$. If a random variable X represents the waiting time for an event to occur in a Poisson process, it is said to follow an **exponential distribution** with rate parameter λ , denoted as $X \sim \text{Exp}(\lambda)$. In this case:

1. The probability density function is

$$\begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

2. $E[X] = \frac{1}{\lambda}$;
3. $\text{Var}(X) = \frac{1}{\lambda^2}$;
4. Memoryless property: for any $s, t \geq 0$, we have $P(X > s + t | X > s) = P(X > t)$;
5. The moment generating function is $M_X(t) = \frac{\lambda}{\lambda - t}$, $t < \lambda$.

Proposition 2.2.3 ► Normal Distribution

Let $\mu \in \mathbb{R}$ and $\sigma^2 > 0$. A random variable X is said to follow a **normal distribution** (or **Gaussian distribution**) with parameters (μ, σ^2) , denoted as $X \sim N(\mu, \sigma^2)$. In this case:

1. The probability density function is $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $x \in \mathbb{R}$;
2. $E[X] = \mu$;
3. $\text{Var}(X) = \sigma^2$;
4. The moment generating function is $M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$;
5. If $X \sim N(\mu, \sigma^2)$, then the standardized variable $Z = \frac{X - \mu}{\sigma}$ follows the **standard normal distribution** $N(0, 1)$;
6. If $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$ and X, Y are independent, then for any constants $a, b \in \mathbb{R}$, $aX + bY \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$.

Proposition 2.2.4 ► Lognormal Distribution

Let Y be a positive random variable. If the new random variable $X = \ln(Y)$ follows a normal distribution with parameters (μ, σ^2) , then Y is said to follow a **lognormal distribution**, denoted as $Y \sim LN(\mu, \sigma^2)$. Note that μ and σ^2 are the mean and variance of the logarithm of the variable, not of the variable itself.

1. The probability density function is

$$f_Y(y) = \begin{cases} \frac{1}{y\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right) & \text{if } y > 0, \\ 0 & \text{otherwise.} \end{cases}$$

2. $E[Y] = e^{\mu + \frac{1}{2}\sigma^2}$;
3. $\text{Var}(Y) = (e^{\sigma^2} - 1)e^{2\mu + \sigma^2}$;
4. The key relationship for problem-solving: If $Y \sim LN(\mu, \sigma^2)$, then

$\ln(Y) \sim N(\mu, \sigma^2)$. This allows us to transform problems about log-normal variables into problems about normal variables.

5. The lognormal distribution arises from the multiplicative version of the Central Limit Theorem. The product of many independent, identically distributed positive random variables will be approximately log-normally distributed.

Proposition 2.2.5 ▶ Gamma Distribution

Let $\alpha > 0$ (shape parameter) and $\lambda > 0$ (rate parameter). A random variable X is said to follow a **Gamma distribution** with parameters (α, λ) , denoted as $X \sim \Gamma(\alpha, \lambda)$. In this case:

1. The probability density function is

$$f(x) = \begin{cases} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} & \text{if } x > 0, \\ 0 & \text{otherwise,} \end{cases}$$

where $\Gamma(\alpha) = \int_0^\infty z^{\alpha-1} e^{-z} dz$ is the Gamma function.

2. $E[X] = \frac{\alpha}{\lambda}$;
3. $\text{Var}(X) = \frac{\alpha}{\lambda^2}$;
4. The moment generating function is $M_X(t) = \left(\frac{\lambda}{\lambda-t}\right)^\alpha$, $t < \lambda$;
5. If $X \sim \Gamma(\alpha_1, \lambda)$, $Y \sim \Gamma(\alpha_2, \lambda)$ and X, Y are independent, then $X + Y \sim \Gamma(\alpha_1 + \alpha_2, \lambda)$;
6. The exponential distribution is a special case: $\text{Exp}(\lambda) \equiv \Gamma(1, \lambda)$;
7. The **Chi-squared distribution** (卡方分布) with k degrees of freedom, denoted χ_k^2 , is a special case: $\chi_k^2 \equiv \Gamma(k/2, 1/2)$.

Proposition 2.2.6 ▶ Central Limit Theorem

Let $X_1, X_2, \dots$ be a sequence of independent and identically distributed (i.i.d.) random variables with finite mean $E[X_i] = \mu$ and finite variance $\text{Var}(X_i) = \sigma^2 > 0$. Let $S_n = \sum_{i=1}^n X_i$. Then the standardized sample mean converges in distribution to a standard normal distribution:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty.$$

Proposition 2.2.7 ▶ Beta Distribution

Let $\alpha > 0$ and $\beta > 0$ be shape parameters. A random variable X is said to follow a **Beta distribution** with parameters (α, β) , denoted as $X \sim \text{Beta}(\alpha, \beta)$, if its support is the interval $(0, 1)$. This distribution is commonly used to model probabilities or proportions.

1. The probability density function is

$$f_X(x) = \begin{cases} \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)} & \text{if } 0 < x < 1, \\ 0 & \text{otherwise,} \end{cases}$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the Beta function.

2. $E[X] = \frac{\alpha}{\alpha+\beta}$;
3. $\text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$;
4. The k -th moment is $E[X^k] = \frac{\Gamma(\alpha+k)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+k)}$ for any $k > -\alpha$;
5. The moment generating function does not have a simple closed form.
6. If $U \sim \Gamma(\alpha, \lambda)$ and $V \sim \Gamma(\beta, \lambda)$ are independent, then the random variable $X = \frac{U}{U+V}$ follows a Beta distribution, $X \sim \text{Beta}(\alpha, \beta)$.

Note.

Beta 分布是在我们相信一个观测结果是由二项分布产生的情况下，导致这个观测结果的 p 的分布， p 是我们认为的二项分布的参数。

Proposition 2.2.8 ▶ Cauchy Distribution

Let $x_0 \in \mathbb{R}$ be the location parameter (median) and $\gamma > 0$ be the scale parameter. A random variable X is said to follow a **Cauchy distribution**, denoted as $X \sim \text{Cauchy}(x_0, \gamma)$. It is a classic example of a distribution for which the moments (like mean and variance) are undefined. The case with $x_0 = 0, \gamma = 1$ is called the **standard Cauchy distribution**.

1. The probability density function is

$$f_X(x) = \frac{1}{\pi\gamma \left[1 + \left(\frac{x-x_0}{\gamma} \right)^2 \right]}, \quad x \in \mathbb{R};$$

2. $E[X]$ is undefined;
3. $\text{Var}(X)$ is undefined;
4. The moment generating function is undefined. However, its characteristic function is well-defined: $\phi_X(t) = E[e^{itX}] = e^{ix_0t - \gamma|t|}$;
5. If $U, V \sim N(0, 1)$ are independent standard normal variables, then their ratio U/V follows the standard Cauchy distribution, $\text{Cauchy}(0, 1)$;
6. The Cauchy distribution is a special case of the Student's t -distribution with one degree of freedom.
7. If $X_1, \dots, X_n$ are i.i.d. from $\text{Cauchy}(x_0, \gamma)$, their sample mean $\bar{X}_n$ has the same distribution, $\bar{X}_n \sim \text{Cauchy}(x_0, \gamma)$. This shows it does not obey the Law of Large Numbers.

Proposition 2.2.9 ▶ Student's t -Distribution

Let $\nu > 0$ be the degrees of freedom. A random variable T is said to follow a **Student's t -distribution** with ν degrees of freedom, denoted as $T \sim t_\nu$. This distribution arises when estimating the mean of a normally distributed population in situations where the sample size is small and the population standard deviation is unknown.

1. The probability density function is

$$f_T(t) = \frac{\Gamma((\nu + 1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad t \in \mathbb{R};$$

2. $E[T] = 0$ for $\nu > 1$. It is undefined for $\nu \leq 1$;
3. $\text{Var}(T) = \frac{\nu}{\nu-2}$ for $\nu > 2$. It is infinite for $1 < \nu \leq 2$ and undefined for $\nu \leq 1$;
4. If $Z \sim N(0, 1)$ and $V \sim \chi_\nu^2$ are independent, then $T = \frac{Z}{\sqrt{V/\nu}} \sim t_\nu$;
5. As $\nu \rightarrow \infty$, the t -distribution converges in distribution to the standard normal distribution $N(0, 1)$.

Proposition 2.2.10 ▶ F-Distribution

Let $d_1 > 0$ and $d_2 > 0$ be the numerator and denominator degrees of freedom, respectively. A random variable X is said to follow an **F-distribution**, denoted as $X \sim F(d_1, d_2)$. It is most commonly used in the analysis of variance (ANOVA).

1. The probability density function is

$$f_X(x) = \begin{cases} \frac{1}{B(d_1/2, d_2/2)} \left(\frac{d_1}{d_2}\right)^{d_1/2} x^{d_1/2-1} \left(1 + \frac{d_1}{d_2}x\right)^{-(d_1+d_2)/2} & \text{if } x > 0, \\ 0 & \text{otherwise;} \end{cases}$$

2. $E[X] = \frac{d_2}{d_2-2}$ for $d_2 > 2$;

3. $\text{Var}(X) = \frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}$ for $d_2 > 4$;

4. If $U_1 \sim \chi_{d_1}^2$ and $U_2 \sim \chi_{d_2}^2$ are independent, then the ratio of the scaled variables $X = \frac{U_1/d_1}{U_2/d_2}$ follows an F-distribution, $X \sim F(d_1, d_2)$;

5. If $T \sim t_{d_2}$, then $T^2 \sim F(1, d_2)$.

2.3 Multivariable Distributions

Proposition 2.3.1 ► Multivariate Normal Distribution

Let $\mu \in \mathbb{R}^k$ be a mean vector and Σ be a $k \times k$ symmetric, positive-definite covariance matrix. A random vector $\mathbf{X} = (X_1, \dots, X_k)^T$ is said to follow a ***multivariate normal distribution***, denoted as $\mathbf{X} \sim N_k(\mu, \Sigma)$.

1. The probability density function is $f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{k/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right)$, $\mathbf{x} \in \mathbb{R}^k$; where $|\Sigma|$ is the determinant of Σ . The term $(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)$ is the Mahalanobis distance squared.
2. $E[\mathbf{X}] = \mu$;
3. $\text{Cov}(\mathbf{X}) = E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)^T] = \Sigma$;
4. The moment generating function is $M_{\mathbf{X}}(\mathbf{t}) = E[e^{\mathbf{t}^T \mathbf{X}}] = e^{\mathbf{t}^T \mu + \frac{1}{2} \mathbf{t}^T \Sigma \mathbf{t}}$;
5. **Marginal Distributions:** Any subset of the components of $\mathbf{X}$ also follows a multivariate normal distribution.
6. **Linear Transformations:** If $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ for a matrix $\mathbf{A}$ of size $m \times k$ and vector $\mathbf{b} \in \mathbb{R}^m$, then $\mathbf{Y} \sim N_m(\mathbf{A}\mu + \mathbf{b}, \mathbf{A}\Sigma\mathbf{A}^T)$.
7. Two components X_i and X_j are independent if and only if their covariance $\Sigma_{ij} = 0$. This is a special and powerful property of the normal distribution.

Proposition 2.3.2 ▶ Multinomial Distribution

Let $n \in \mathbb{N}^+$ be the number of trials, and $\mathbf{p} = (p_1, \dots, p_k)$ be a vector of probabilities such that $p_i \geq 0$ for all i and $\sum_{i=1}^k p_i = 1$. If a random vector $\mathbf{X} = (X_1, \dots, X_k)$ represents the number of times each of the k possible outcomes occurs in n independent trials, then $\mathbf{X}$ follows a **multinomial distribution**, denoted as $\mathbf{X} \sim \text{Mult}(n, \mathbf{p})$.

1. The probability mass function is $P(X_1 = n_1, \dots, X_k = n_k) = \frac{n!}{n_1! n_2! \dots n_k!} p_1^{n_1} p_2^{n_2} \dots p_k^{n_k}$, where $n_i \in \{0, 1, \dots, n\}$ and $\sum_{i=1}^k n_i = n$.
2. $E[X_i] = np_i$;
3. $\text{Var}(X_i) = np_i(1 - p_i)$;
4. $\text{Cov}(X_i, X_j) = -np_i p_j$ for $i \neq j$. The negative covariance reflects the constraint that $\sum X_i = n$.
5. The binomial distribution is a special case where $k = 2$.

Proposition 2.3.3 ▶ Dirichlet Distribution

Let $\alpha = (\alpha_1, \dots, \alpha_k)$ be a vector of concentration parameters with $\alpha_i > 0$. A random vector $\mathbf{X} = (X_1, \dots, X_k)$ is said to follow a **Dirichlet distribution**, denoted as $\mathbf{X} \sim \text{Dir}(\alpha)$. Its support is the standard $(k - 1)$ -simplex, i.e., $S_{k-1} = \{\mathbf{x} \in \mathbb{R}^k : x_i > 0, \sum_{i=1}^k x_i = 1\}$. It is the multivariate generalization of the Beta distribution.

1. The probability density function is $f(\mathbf{x}; \alpha) = \frac{1}{B(\alpha)} \prod_{i=1}^k x_i^{\alpha_i - 1}$, where $B(\alpha) = \frac{\prod_{i=1}^k \Gamma(\alpha_i)}{\Gamma(\sum_{i=1}^k \alpha_i)}$ is the multivariate Beta function.
2. Let $\alpha_0 = \sum_{i=1}^k \alpha_i$.
3. $E[X_i] = \frac{\alpha_i}{\alpha_0}$;
4. $\text{Var}(X_i) = \frac{\alpha_i(\alpha_0 - \alpha_i)}{\alpha_0^2(\alpha_0 + 1)}$;
5. $\text{Cov}(X_i, X_j) = \frac{-\alpha_i \alpha_j}{\alpha_0^2(\alpha_0 + 1)}$ for $i \neq j$;
6. The Dirichlet distribution is the conjugate prior for the parameter vector $\mathbf{p}$ of the multinomial distribution.

Proposition 2.3.4 ► Wishart Distribution

Let $\mathbf{V}$ be a $p \times p$ symmetric positive-definite scale matrix and let $n \geq p$ be the degrees of freedom. A random matrix $\mathbf{W}$ is said to follow a **Wishart distribution**, denoted $\mathbf{W} \sim W_p(n, \mathbf{V})$, if it is the distribution of the sample covariance matrix. Specifically, if $\mathbf{X}_1, \dots, \mathbf{X}_n$ are i.i.d. samples from $N_p(\mathbf{0}, \mathbf{V})$, then $\mathbf{W} = \sum_{i=1}^n \mathbf{X}_i \mathbf{X}_i^T \sim W_p(n, \mathbf{V})$.

1. The probability density function for a symmetric positive-definite matrix $\mathbf{X}$ is $f(\mathbf{X}) = \frac{|\mathbf{X}|^{(n-p-1)/2} \exp\left(-\frac{1}{2}\text{tr}(\mathbf{V}^{-1}\mathbf{X})\right)}{2^{np/2} |\mathbf{V}|^{n/2} \Gamma_p(n/2)}$, where $\Gamma_p(\cdot)$ is the multivariate Gamma function.
2. $E[\mathbf{W}] = n\mathbf{V}$;
3. The Wishart distribution is the multivariate generalization of the Chi-squared distribution. If $p = 1$ and $\mathbf{V} = 1$, then $W_1(n, 1)$ is the χ_n^2 distribution.
4. **Additive Property:** If $\mathbf{W}_1 \sim W_p(n_1, \mathbf{V})$ and $\mathbf{W}_2 \sim W_p(n_2, \mathbf{V})$ are independent, then $\mathbf{W}_1 + \mathbf{W}_2 \sim W_p(n_1 + n_2, \mathbf{V})$.

Proposition 2.3.5 ► Uniform Distribution on the Sphere

This distribution describes a point chosen at random from the surface of a unit sphere in $\mathbb{R}^n$. Let $S^{n-1} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\|_2 = 1\}$ be the unit sphere. A random vector $\mathbf{X}$ is uniformly distributed on S^{n-1} . This is a foundational example of a distribution on a Riemannian manifold.

1. The probability density function is constant with respect to the surface area measure (the canonical volume form on the sphere). If A_{n-1} is the surface area of S^{n-1} , the density is $f(\mathbf{x}) = 1/A_{n-1}$.
2. The surface area of S^{n-1} is given by $A_{n-1} = \frac{2\pi^{n/2}}{\Gamma(n/2)}$.
3. **Generation:** If $\mathbf{Z} = (Z_1, \dots, Z_n)^T$ where $Z_i \sim N(0, 1)$ are i.i.d., then the vector $\mathbf{X} = \frac{\mathbf{Z}}{\|\mathbf{Z}\|_2}$ is uniformly distributed on S^{n-1} . This is a consequence of the rotational symmetry of the multivariate normal distribution.
4. $E[\mathbf{X}] = \mathbf{0}$ due to symmetry.
5. $\text{Cov}(\mathbf{X}) = E[\mathbf{X}\mathbf{X}^T] = \frac{1}{n}\mathbf{I}_n$, where $\mathbf{I}_n$ is the $n \times n$ identity matrix.

Law of Large Number

3.1 Independence

Definition 3.1.1 ► Independence

- Two events A and B are **independent** if $P(A \cap B) = P(A)P(B)$.
- Two random variables X and Y are **independent** if for all $C, D \in \mathcal{R}$,

$$P(X \in C, Y \in D) = P(X \in C)P(Y \in D)$$

that is, the events $A = \{X \in C\}$ and $B = \{Y \in D\}$ are independent.

- Two σ -fields $\mathcal{F}$ and $\mathcal{G}$ are **independent** if for all $A \in \mathcal{F}$ and $B \in \mathcal{G}$ the events A and B are independent.

Exercise 3.1.2

1. Show that if A and B are independent, then so are A^c and B , A and B^c , and A^c and B^c .
2. Conclude that events A and B are independent if and only if their indicator random variables 1_A and 1_B are independent.

Proof. 1. If A and B are independent, then

$$P(A \cap B) = P(A)P(B)$$

Note that

$$P(A \cap B) + P(A^c \cap B) = P(B)$$

We have

$$P(A^c \cap B) = P(B)(1 - P(A)) = P(B)P(A^c)$$

Similarly,

$$P(A \cap B^c) = P(A)P(B^c)$$

$$P(A^c \cap B^c) = P(B^c) - P(A \cap B^c) = P(B^c) - P(A)P(B^c) = P(A^c)P(B^c)$$

2.

$$\sigma(1_A) = \{\emptyset, A, A^c, \Omega\}, \quad \sigma(1_B) = \{\emptyset, B, B^c, \Omega\}$$

If $\sigma(1_A)$ and $\sigma(1_B)$ are independent, it is clear that A and B are as well. Conversely, only need to show that $\emptyset$ and A are independent, the remaining follows from 1. and symmetry. We have

$$P(\emptyset \cap A) = 0 = P(\emptyset)P(A)$$

□

Definition 3.1.3

- σ -fields $\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_n$ are **independent** if whenever $A_i \in \mathcal{F}_i$ for $i = 1, \dots, n$, we have

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i)$$

- Random variables $X_1, \dots, X_n$ are **independent** if whenever $B_i \in \mathcal{R}$ for $i = 1, \dots, n$ we have

$$P\left(\bigcap_{i=1}^n \{X_i \in B_i\}\right) = \prod_{i=1}^n P(X_i \in B_i)$$

- Sets $A_1, \dots, A_n$ are **independent** if whenever $I \subset \{1, \dots, n\}$ we have

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

Remark.

The last definition looks slightly different from the former two, but it is consistent with the independence of the indicator variables 1_{A_i} , $1 \leq i \leq n$.

Exercise 3.1.4

Let $A_1, A_2, \dots, A_n$ be independent. Show

1. $A_1^c, A_2, \dots, A_n$ are independent;
2. $1_{A_1}, \dots, 1_{A_n}$ are independent.

Proof. 1. If $1 \notin I$, then it is obvious that

$$P(\cap_{i \in I} A_i) = \prod_{i \in I} P(A_i).$$

If $1 \in I$, note that

$$\begin{aligned} P(\cap_{i \in I \setminus \{1\}} A_i \cap A_1^c) &= P(\cap_{i \in I \setminus \{1\}} A_i) - P(\cap_{i \in I \setminus \{1\}} A_i \cap A_1) \\ &= \prod_{i \in I \setminus \{1\}} P(A_i) - \prod_{i \in I} P(A_i) \\ &= (1 - P(A_1)) \prod_{i \in I \setminus \{1\}} P(A_i) \\ &= P(A_1^c) \prod_{i \in I \setminus \{1\}} P(A_i) \end{aligned}$$

2.

$$\sigma(1_{A_i}) = \{\emptyset, A_i^c, A_i, \Omega\}$$

Take $B_i \in \sigma(1_{A_i})$, $i = 1, \dots, n$. $B_i = \emptyset$ for some i , then both $P(\cap_{i=1}^n B_i)$ and $\prod_{i=1}^n P(B_i)$ are zero. The remainings follows from 1.

□

Example 3.1.5

Let X_1, X_2, X_3 be independent random variables with

$$P(X_i = 0) = P(X_i = 1) = 1/2$$

Let $A_1 = \{X_2 = X_3\}$, $A_2 = \{X_3 = X_1\}$ and $A_3 = \{X_1 = X_2\}$. These events are pairwise independent since if $i \neq j$, then

$$P(A_i \cap A_j) = P(X_1 = X_2 = X_3) = 1/4 = P(A_i)P(A_j)$$

but they are not independent since

$$P(A_1 \cap A_2 \cap A_3) = 1/4 \neq 1/8 = P(A_1)P(A_2)P(A_3)$$

3.1.1 Sufficient Conditions

Definition 3.1.6 ▶ Independence between collections

Collections of sets $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n \subset \mathcal{F}$ are said to be **independent** if whenever $A_i \in \mathcal{A}_i$ and $I \subset \{1, \dots, n\}$ we have $P(\bigcap_{i \in I} A_i) = \prod_{i \in I} P(A_i)$.

Lemma 3.1.7

Without loss of generality we can suppose each $\mathcal{A}_i$ contains Ω . In this case the condition is equivalent to

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i) \text{ whenever } A_i \in \mathcal{A}_i$$

since we can set $A_i = \Omega$ for $i \notin I$.

Definition 3.1.8 ▶ π, λ -systems

- We say that $\mathcal{A}$ is a **π -system** if it is closed under intersection, that is, if $A, B \in \mathcal{A}$ then $A \cap B \in \mathcal{A}$.
- We say that $\mathcal{L}$ is a **λ -system** if
 - (i) $\Omega \in \mathcal{L}$.
 - (ii) If $A, B \in \mathcal{L}$ and $A \subset B$, then $B - A \in \mathcal{L}$.
 - (iii) If $A_n \in \mathcal{L}$ and $A_n \uparrow A$, then $A \in \mathcal{L}$.

Theorem 3.1.9 ▶ $\pi - \lambda$ Theorem

If $\mathcal{P}$ is a π -system and $\mathcal{L}$ is a λ -system that contains $\mathcal{P}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.

Theorem 3.1.10

Suppose $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent and each $\mathcal{A}_i$ is a π -system. Then $\sigma(\mathcal{A}_1), \sigma(\mathcal{A}_2), \dots, \sigma(\mathcal{A}_n)$ are independent.

Proof sketch.

对于任一取定的事件 F , “(取定任一集族中) 与 F 独立” 的性质集是一个 λ -system. 由于 $\mathcal{A}_1$ 是 π -system, 于是 $\mathcal{A}_1$ 中与 F 独立的集合包含了 $\sigma(\mathcal{A}_1)$, 通过任意的 F , 传递归纳集族的独立性一步步“升级”到它们的生成 σ -代数上.

Remark.

It is not easy to show that if $A, B \in \mathcal{L}$ then $A \cap B \in \mathcal{L}$ or $A \cup B \in \mathcal{L}$, but it is easy to check that if $A, B \in \mathcal{L}$ with $A \subseteq B$, then $B - A \in \mathcal{L}$.

Proof. Let $F \in \mathcal{P}$ be any event. We define

$$\mathcal{L} := \{A \in \mathcal{A}_1 : P(A \cap F) = P(A)P(F)\}$$

We check that $\mathcal{L}$ is a λ -system. (i) follows from $P(A \cap \Omega) = P(A) \cdot 1 = P(A)P(\Omega)$. To show (ii), we take $A, B \in \mathcal{L}, B \supseteq A$. Note that

$$(B - A) \cap F = (B \cap F) - (A \cap F),$$

we have

$$\begin{aligned} P((B \setminus A) \cap F) &= P((B \cap F) - (A \cap F)) \\ &= P(B \cap F) - P(A \cap F) \\ &= P(B)P(F) - P(A)P(F) \\ &= (P(B) - P(A))P(F) = P((B - A))P(F) \end{aligned}$$

which shows that $(B - A) \in \mathcal{L}$

For (iii), take $B_n \in \mathcal{L}, B_n \uparrow B$. Then since $(B_n \cap F) \uparrow B \cap F$, we have

$$\begin{aligned} P(B \cap F) &= \lim_{n \rightarrow \infty} P(B_n \cap F) \\ &= \lim_{n \rightarrow \infty} P(B_n)P(F), \\ &= P(B)P(F) \end{aligned}$$

which implies that $B \in \mathcal{L}$. Thus $\mathcal{L}$ is a λ -system. It follows from $\pi - \lambda$ Theorem that $\mathcal{L} \supseteq \sigma(\mathcal{A}_1)$ For each $A_2, \dots, A_n, A_i \in \mathcal{A}_i, i = 2, 3, \dots, n$, by taking $F = \bigcap_{i=2}^n A_i$, we have

$$P(A \cap A_2 \cap \dots \cap A_n) = P(A)P\left(\bigcap_{i=2}^n A_i\right) = P(A)P(A_2) \dots P(A_n), \quad \forall A \in \sigma(\mathcal{A}_1)$$

We get that

if $\mathcal{A}_1, \dots, \mathcal{A}_n$ are independent π -systems, then $\sigma(\mathcal{A}_1), \mathcal{A}_2, \dots, \mathcal{A}_n$ are as well (*)

By applying (*) into $\mathcal{A}_2, \dots, \mathcal{A}_n, \sigma(\mathcal{A}_1)$, we get $\sigma(\mathcal{A}_2), \mathcal{A}_3, \dots, \mathcal{A}_n, \sigma(\mathcal{A}_1)$ are

independent. The proof completes after n iterations. $\square$

Theorem 3.1.11

In order for $X_1, \dots, X_n$ to be independent, it is sufficient that for all $x_1, \dots, x_n \in (-\infty, \infty]$

$$P(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_{i=1}^n P(X_i \leq x_i)$$

Proof. Let $\mathcal{A}_i$ be the collection of the sets of the form $\{X_i \leq x_i\}$. Since

$$\{X_i \leq x\} \cap \{X_i \leq y\} = \{X_i \leq x \wedge y\}$$

$\mathcal{A}_i$ is a π -system. Since we allowed $x_i = \infty$, $\Omega \in \mathcal{A}_i$. Exercise 1.3.10 implies that $\sigma(\mathcal{A}_i) = \sigma(X_i)$, so the result follows from Theorem 3.1.11. $\square$

Theorem 3.1.12

Suppose $\mathcal{F}_{i,j}$, $1 \leq i \leq n$, $1 \leq j \leq m(i)$ are independent π -systems and let $\mathcal{G}_i = \sigma(\bigcup_j \mathcal{F}_{i,j})$. Then $\mathcal{G}_1, \dots, \mathcal{G}_n$ are independent.

Note.

对于独立的 π -systems, 可以把它们按任意分组方式, “攢起来” 做 σ -代数扩张, 得到新的独立系统.

Proof. Let $\mathcal{A}_i$ be the collection of sets of the form $\bigcap_j A_{i,j}$ where $A_{i,j} \in \mathcal{F}_{i,j}$. $\mathcal{A}_i$ is a π -system that contains Ω and contains $\bigcup_j \mathcal{F}_{i,j}$, so Theorem 3.1.10 implies $\sigma(\mathcal{A}_i) = \mathcal{G}_i$ are independent. $\square$

Theorem 3.1.13

If for $1 \leq i \leq n$, $1 \leq j \leq m(i)$, $X_{i,j}$ are independent and $f_i : \mathbf{R}^{m(i)} \rightarrow \mathbf{R}$ are measurable, then $f_i(X_{i,1}, \dots, X_{i,m(i)})$ are independent.

Proof. $\sigma(X_{i,j})$, $1 \leq i \leq n$, $1 \leq j \leq m(i)$ are independent π -systems. Then

$\mathcal{G}_i = \sigma\left(\bigcup_j (\sigma(X_{i,j}))\right)$ are independent. Take any Borel set $B \subseteq \mathbb{R}$, then

$$\begin{aligned} (f_i(X_{i,1}, \dots, X_{i,m(i)}))^{-1}(B) &= (X_{i,1}, \dots, X_{i,m(i)})^{-1}(f_i^{-1}(B)) \\ &\subseteq \sigma\left(\bigcup_j (\sigma(X_{i,j}))\right) \end{aligned}$$

Thus $f_i(X_{i,1}, \dots, X_{i,m(i)})$ are independent. □

3.1.2 Independence, Distribution, and Expectation

Theorem 3.1.14

Suppose $X_1, \dots, X_n$ are independent random variables and X_i has distribution μ_i . Then $(X_1, \dots, X_n)$ has distribution $\mu_1 \times \dots \times \mu_n$.

Proof.

$$\begin{aligned} P((X_1, \dots, X_n) \in A_1 \times \dots \times A_n) &= P(X_1 \in A_1, \dots, X_n \in A_n) \\ &= \prod_{i=1}^n P(X_i \in A_i) \\ &= \prod_{i=1}^n \mu_i(A_i) \\ &= (\mu_1 \times \dots \times \mu_n)(A_1 \times \dots \times A_n) \end{aligned}$$

The above shows that the distribution of $(X_1, \dots, X_n)$ and the measure $\mu_1 \times \dots \times \mu_n$ agree on the sets of the form $A_1 \times \dots \times A_n$, which forms a π -system generates $\mathcal{R}^n$. Since the collection of the sets on which two measures here agree forms a $\mathcal{L}$ -system. Then it follows from Theorem 3.1.9 that $\mu_1 \times \dots \times \mu_n$ and $(X_1, \dots, X_n)$ agree on $\mathcal{R}^n$. □

Theorem 3.1.15

Suppose X and Y are independent and have distributions μ and ν . If $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a measurable function with $h \geq 0$ or $E|h(X, Y)| < \infty$, then

$$Eh(X, Y) = \iint h(x, y)\mu(dx)\nu(dy)$$

In particular, if $h(x, y) = f(x)g(y)$ where $f, g : \mathbf{R} \rightarrow \mathbf{R}$ are measurable functions with $f, g \geq 0$ or $E|f(X)|$ and $E|g(Y)| < \infty$, then

$$Ef(X)g(Y) = Ef(X) \cdot Eg(Y)$$

Proof. From the Changing of Variables and the Fubini's Theorem, we have

$$Eh(X, Y) = \int h \, d(\mu \times \nu) = \iint h(x, y) \mu(dx) \nu(dy)$$

To show the second part, we first assume that $f, g \geq 0$, then

$$\begin{aligned} Ef(X)g(Y) &= \iint f(x)g(y) \mu(dx) \nu(dy) \\ &= \int g(y) \int f(x) \mu(dx) \nu(dy) \\ &= \int g(y) Ef(X) \nu(dy) \\ &= Ef(X) Eg(Y) \end{aligned}$$

For another assumption, we apply the above result to the nonnegative function $|f|, |g|$, to get $E|h(X, Y)| < \infty$, and we repeat the last argument to prove the desired result. $\square$

Corollary 3.1.16

If $X_1, \dots, X_n$ are independent and have

- (a) $X_i \geq 0$ for all i , or
- (b) $E|X_i| < \infty$ for all i , then

$$E\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n EX_i$$

that is, the expectation on the left exists and has the value given on the right.

3.1.3 Sums of Independent Random Variables

Theorem 3.1.17

If X and Y are independent, $F(x) = P(X \leq x)$, and $G(y) = P(Y \leq y)$, then

$$P(X + Y \leq z) = \int F(z - y) dG(y)$$

The integral on the right-hand side is called the **convolution** of F and G and is denoted $F * G(z)$. The meaning of $dG(y)$ will be explained in the proof.

Proof.

$$\begin{aligned} P(X + Y \leq z) &= \int 1_{(x+y \leq z)} d(\mu \times \nu) \\ &= \int \int 1_{(x+y \leq z)} d\mu d\nu \\ &= \int F(z - y) d\nu \\ &= \int F(z - y) dG(y) \end{aligned}$$

where $dG(y)$ is just a notation for $d\nu$. We apply this notation motivated from the Lebesgue-Stieljes integration. $\square$

Theorem 3.1.18

Suppose that X with density f and Y with distribution function G are independent. Then $X + Y$ has density

$$h(x) = \int f(x - y) dG(y)$$

When Y has density g , the last formula can be written as

$$h(x) = \int f(x - y) g(y) dy$$

Proof.

$$F(x) = \int_{-\infty}^x f(x) dx$$

Since everything is nonnegative, from Fubini's theorem, we get

$$\begin{aligned} P(X + Y \leq z) &= \int \int_{-\infty}^{z-y} f(x) dx dG(y) \\ &= \int \int_{-\infty}^z f(x-y) dx dG(y) \\ &= \int_{-\infty}^z \int f(x-y) dG(y) dx \\ &= \int_{-\infty}^z h(x) dx \end{aligned}$$

which shows that $X + Y$ has density $h(x)$.

When Y has density g , then

$$\int f(x-y) dG(y) = \int f(x-y) g(y) dy$$

□

Example 3.1.19 ► The gamma density

The **gamma density** with parameters α and λ is given by

$$f(x) = \begin{cases} \lambda^\alpha x^{\alpha-1} e^{-\lambda x} / \Gamma(\alpha) & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$.

Theorem 3.1.20

If $X = \text{gamma}(\alpha, \lambda)$ and $Y = \text{gamma}(\beta, \lambda)$ are independent, then $X + Y$ is $\text{gamma}(\alpha + \beta, \lambda)$. Consequently if $X_1, \dots, X_n$ are independent exponential (λ) r.v's, then $X_1 + \dots + X_n$ has a $\text{gamma}(n, \lambda)$ distribution.

Example 3.1.21 ▶ Normal distribution

In Example 1.6.2, we introduced the normal density with mean μ and variance a ,

$$(2\pi a)^{-1/2} \exp(-(x - \mu)^2/2a).$$

Theorem 3.1.22

If $X = \text{normal}(\mu, a)$ and $Y = \text{normal}(\nu, b)$ are independent, then $X + Y = \text{normal}(\mu + \nu, a + b)$.

3.2 Weak Laws of Large Numbers

Random Vector and its Distribution

4.1 The additivity of some distributions

Example 4.1.1 ► The additivity of Poisson Distribution

Suppose that $X \sim P(\lambda_1)$, $Y \sim P(\lambda_2)$, and X, Y are independent. Then

$$Z = X + Y \sim P(\lambda_1 + \lambda_2)$$

Example 4.1.2 ► The additivity of Binomial Distribution

Suppose that $X \sim b(n, p)$, $Y \sim b(m, p)$ and X, Y are independent. Then

$$Z = X + Y \sim b(n + m, p)$$

Example 4.1.3 ► the additivity of Normal Distribution

Suppose that $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$ and X, Y are independent. Then

$$Z = X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Example 4.1.4 ► the additivity of Gamma Distribution

Suppose that $X \sim Ga(\alpha_1, \lambda)$, $Y_2 \sim Ga(\alpha_2, \lambda)$, and X, Y are independent. Then

$$Z = X + Y \sim Ga(\alpha_1 + \alpha_2, \lambda)$$

Characteristic Functions

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